GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Lab: Deep Learning

DS-563-MJP

Solved Practical Manual (TensorFlow / Keras)

Semester II · Lab (MJP)

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This solved practical manual covers the Deep Learning lab assignments from the official SPPU M.Sc. Data Science syllabus. Each practical states the aim followed by a complete, runnable solution and the expected output. Use it as a reference — type the code yourself and experiment with the parameters to learn effectively.

Requirements: Python 3.x with numpy, pandas, matplotlib, scikit-learn, scipy installed (pip install numpy pandas matplotlib scikit-learn scipy).

Requirements: tensorflow (pip install tensorflow).

Unit 1: Feedforward network on the XOR problem

```
import numpy as np, tensorflow as tf
X = np.array([[0,0],[0,1],[1,0],[1,1]]); y = np.array([0,1,1,0])
model = tf.keras.Sequential([
    tf.keras.layers.Dense(4, activation="relu", input_shape=(2,)),
    tf.keras.layers.Dense(1, activation="sigmoid")])
model.compile(optimizer="adam", loss="binary_crossentropy", metrics=["accuracy"])
model.fit(X, y, epochs=500, verbose=0)
print("Predictions:", model.predict(X).round())
```

Unit 2: MLP classifier on MNIST

```
import tensorflow as tf
(xtr,ytr),(xte,yte) = tf.keras.datasets.mnist.load_data()
xtr, xte = xtr/255.0, xte/255.0
model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=(28,28)),
    tf.keras.layers.Dense(128, activation="relu"),
    tf.keras.layers.Dense(10, activation="softmax")])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
metrics=["accuracy"])
model.fit(xtr, ytr, epochs=3, validation_split=0.1)
print("Test accuracy:", model.evaluate(xte, yte, verbose=0)[1])
```

Unit 3: CNN on CIFAR-10

```
import tensorflow as tf
(xtr,ytr),(xte,yte) = tf.keras.datasets.cifar10.load_data()
xtr, xte = xtr/255.0, xte/255.0
model = tf.keras.Sequential([
    tf.keras.layers.Conv2D(32,3,activation="relu",input_shape=(32,32,3)),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Conv2D(64,3,activation="relu"),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(64,activation="relu"),
    tf.keras.layers.Dense(10,activation="softmax")])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
metrics=["accuracy"])
model.fit(xtr, ytr, epochs=5, validation_split=0.1)
```

Unit 4: RNN / LSTM sentiment analysis (IMDB)

```
import tensorflow as tf
(xtr,ytr),(xte,yte) = tf.keras.datasets.imdb.load_data(num_words=10000)
xtr = tf.keras.preprocessing.sequence.pad_sequences(xtr, maxlen=200)
xte = tf.keras.preprocessing.sequence.pad_sequences(xte, maxlen=200)
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(10000, 32),
    tf.keras.layers.LSTM(32),
    tf.keras.layers.Dense(1, activation="sigmoid")])
model.compile(optimizer="adam", loss="binary_crossentropy", metrics=["accuracy"])
model.fit(xtr, ytr, epochs=2, validation_split=0.2)
```

Unit 5: Ethical considerations — explainability demo

Aim: Interpret a model's predictions to address fairness/transparency (Unit 5).

```
# Feature importance as a simple explainability tool
from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import load_breast_cancer
X, y = load_breast_cancer(return_X_y=True)
rf = RandomForestClassifier().fit(X, y)
import numpy as np
top = np.argsort(rf.feature_importances_)[::-1][:5]
print("Top-5 influential features (indices):", top)
```

Discuss: differential privacy, bias/fairness and accountability as required by the syllabus.